

Ans. to the Ques. no-1

a) From the table,

$$P(X=2, Y=2) = 0.20$$

b) From the table,

$$\begin{aligned} P(X=1) &= P(X=1, Y=0) + P(X=1, Y=1) + P(X=1, Y=2) \\ &= 0.06 + 0.24 + 0.10 \\ &= 0.40 \end{aligned}$$

$$\begin{aligned} P(Y=1) &= P(X=0, Y=1) + P(X=1, Y=1) + P(X=2, Y=1) \\ &= 0.04 + 0.24 + 0.14 \\ &= 0.42 \end{aligned}$$

c) For independency,

$$P(X, Y) = P(X) \cdot P(Y)$$

When $X=Y=1$,

$$P(X, Y) = P(X=1, Y=1) = 0.24$$

$$P(X) \cdot P(Y) = P(X=1) \cdot P(Y=1) = 0.40 \times 0.42 = 0.168$$

$$\text{So, } P(X=1, Y=1) \neq P(X=1) \cdot P(Y=1)$$

So, X and Y are not independent.

Ans. to the Question no-2

2a)

	Light	Dark
Chrome	$\frac{45}{150} = 0.3$	$\frac{35}{150} = 0.23$
Firefox	$\frac{30}{150} = 0.2$	$\frac{40}{150} = 0.267$

Users using Light theme 11,
 $= 45 + 30 = 75$

b)

From the table, $P(\text{Chrome and Dark}) = 0.23$

c)

$$\text{From the table, } P(\text{Chrome}) = \frac{45+35}{150} = \frac{80}{150} = 0.533$$

$$P(\text{Dark}) = \frac{35+40}{150} = \frac{75}{150} = 0.50$$

d) To be independent,

$$P(\text{Browser, Theme}) = P(\text{Browser}) \times P(\text{Theme})$$

From the table,

$$P(\text{Browser, Theme})$$

$$P(\text{Browser} = \text{Chrome, Theme} = \text{Dark}) = 0.23$$

$$P(\text{Browser} = \text{Chrome}) \times P(\text{Theme} = \text{Dark}) = 0.533 \times 0.50 \\ = 0.2665$$

$$\therefore P(\text{Browser, Theme}) \neq P(\text{Browser}) \times P(\text{Theme})$$

e) We know,

$$P(\text{Light} | \text{Firefox}) = \frac{P(\text{Firefox and Light})}{P(\text{Firefox})} \\ = \frac{0.2}{\frac{30+40}{150}} = 0.429$$

Ans. to the Question no-3

a) Since A and B are conditionally independent given C,

$$P(A \cap B | C) = P(A | C) \cdot P(B | C) \cdot P(C) \\ = 0.5 \times 0.6 \times 0.25 \\ = 0.075$$
$$\left\{ \begin{array}{l} P(A | C) = 0.5 \\ P(B | C) = 0.6 \\ P(C) = 0.25 \end{array} \right.$$

b) Using law of total probability,

$$P(A \cap B) = P(A \cap B | C) \cdot P(C) + P(A \cap B | \neg C) \cdot P(\neg C) \\ = P(A | C) \cdot P(B | C) \cdot P(C) + P(A | \neg C) \cdot P(B | \neg C) \cdot \{1 - P(C)\} \\ = 0.5 \times 0.6 \times 0.25 + 0.2 \times 0.3 \times (1 - 0.25) \\ = 0.12$$

Ans. to the Ques. no-4

a) $P(\text{red}_1, \text{red}_2) = \frac{5}{8} \times \frac{4}{7} = 0.357$

b)
$$P(\text{red}_1 | \text{red}_2) = \frac{P(\text{red}_1, \text{red}_2)}{P(\text{red}_2)}$$
$$= \frac{\frac{5}{8} \times \frac{4}{7}}{P(\text{red}_1, \text{red}_2) + P(\text{black}_1, \text{red}_2)}$$
$$= \frac{\frac{5}{8} \times \frac{4}{7}}{\frac{5}{8} \times \frac{4}{7} + \frac{3}{8} \times \frac{5}{7}} = \frac{4}{7}$$

Ans. to the Question no-5

a) $P(s) = 0.4$ $P(\neg s) = 0.6$
 $P(L|s) = 0.7$ $P(U|s) = 0.5$
 $P(L|\neg s) = 0.2$ $P(U|\neg s) = 0.1$

$\therefore P(L \cap U | s) = \frac{P(L \cap U | s) \cdot P(s)}{P(L \cap U)}$
$$= \frac{P(L|s) \cdot P(U|s) \cdot P(s)}{P(L \cap U | s) \cdot P(s) + P(L \cap U | \neg s) \cdot P(\neg s)}$$
$$= \frac{P(L|s) \cdot P(U|s) \cdot P(s)}{P(L|s) \cdot P(U|s) \cdot P(s) + P(L|\neg s) \cdot P(U|\neg s) \cdot P(\neg s)}$$
$$= \frac{0.7 \times 0.5 \times 0.4}{0.7 \times 0.5 \times 0.4 + 0.2 \times 0.1 \times 0.6}$$
$$= \frac{35}{38} \approx 0.921$$

b) $P(s | L \cap U) = 0.921$
 $\therefore P(\neg s | L \cap U) = 1 - 0.921 = 0.079$

MAP assigns that maximizes the posterior probability.

So, This message will be classified as Spam.

Ans. to the Ques. no-6

a) Using Bayes' Theorem,

$$\begin{aligned}P(A|D) &= \frac{P(D|A) P(A)}{P(D)} \\&= \frac{P(D|A) P(A)}{P(D|A) P(A) + P(D|B) P(B)} \\&= \frac{0.02 \times 0.65}{0.02 \times 0.65 + 0.06 \times 0.35} \\&= \frac{13}{34} \approx 0.382 \approx 38.2\%\end{aligned}$$

$$\begin{aligned}P(A) &= 0.65 \\P(D|A) &= 0.02 \\P(B) &= 0.35 \\P(D|B) &= 0.06\end{aligned}$$

b) Using Bayes' Theorem,

$$\begin{aligned}P(B|D) &= \frac{P(D|B) P(B)}{P(D)} \\&= \frac{0.06 \times 0.35}{0.02 \times 0.65 + 0.06 \times 0.35} \\&= \frac{21}{34} \approx 0.618 \approx 61.8\%\end{aligned}$$

So, we see production Line B is more likely.

Ans. to the Ques no-7

$$\begin{aligned}\underline{a)} \quad P(\text{Tab} | S \cap T \cap C) &= \frac{P(S \cap T \cap C | \text{Tab}) P(\text{Tab})}{P(S \cap T \cap C)} \\&= \frac{P(S | \text{Tab}) \cdot P(T | \text{Tab}) \cdot P(C | \text{Tab}) \cdot P(\text{Tab})}{P(S \cap T \cap C | \text{Tab}) \cdot P(\text{Tab}) + P(S \cap T \cap C | \text{Ph}) \cdot P(\text{Ph})} \\&= \frac{P(S | \text{Tab}) \cdot P(T | \text{Tab}) \cdot P(C | \text{Tab}) \cdot P(\text{Tab})}{P(S | \text{Tab}) \cdot P(T | \text{Tab}) \cdot P(C | \text{Tab}) \cdot P(\text{Tab}) + P(S | \text{Ph}) \cdot P(T | \text{Ph}) \cdot P(C | \text{Ph}) \cdot P(\text{Ph})} \\&= \frac{0.8 \times 0.6 \times 0.4 \times 0.3}{0.8 \times 0.6 \times 0.4 \times 0.3 + 0.3 \times 0.1 \times 0.9 \times 0.7} \\&= \frac{64}{73} \approx 0.753 \approx 75.3\%\end{aligned}$$

$$\begin{aligned}P(\text{Ph}) &= 0.7 \\P(\text{Tab}) &= 0.3 \\P(S | \text{Ph}) &= 0.3 \\P(T | \text{Ph}) &= 0.1 \\P(C | \text{Ph}) &= 0.9 \\P(S | \text{Tab}) &= 0.8 \\P(T | \text{Tab}) &= 0.6 \\P(C | \text{Tab}) &= 0.4\end{aligned}$$

$$\begin{aligned}
 \underline{b)} \quad P(Ph|sNTnc) &= \frac{P(sNTnc|\text{Tablet}) P(Ph)}{P(sNTnc)} \\
 &= \frac{P(s|Ph) \cdot P(T|Ph) P(C|Ph) \cdot P(Ph)}{P(sNTnc)} \\
 &= \frac{0.3 \times 0.1 \times 0.9 \times 0.7}{0.8 \times 0.6 \times 0.4 \times 0.3 + 0.3 \times 0.1 \times 0.9 \times 0.7} \\
 &= \frac{21}{85} \approx 0.247 \approx 24.7\%
 \end{aligned}$$

So, we see that the classifier will assign the device as Tablet.

Ans. to the Ques. no-8

a)

ID	Income	Discretized	Class
1	85	High	Approved
2	38	Low	Denied
3	90	High	Approved
4	50	Med	Denied
5	72	High	Approved
6	32	Low	Denied
7	42	Med	Denied
8	75	High	Approved
9	68	Med	Approved
10	35	Low	Denied
11	65	Med	Approved
12	55	Med	Approved
13	46	Med	Denied
14	28	Low	Denied
15	33	Low	Denied

Low class : Approved : 0
Denied : 5

Med class : Approved : 3
Denied : 3

High class : Approved : 4
Denied : 0

b) $P(\text{Approved}) = \frac{0+3+4}{15} = \frac{7}{15} = 0.467$
 $P(\text{Denied}) = \frac{5+3+0}{15} = \frac{8}{15} = 0.533$

c) Discretized Income:

$$P(\text{Low} | \text{Approved}) = \frac{0}{7} = 0$$

$$P(\text{Med} | \text{Approved}) = \frac{3}{7} = 0.429$$

$$P(\text{High} | \text{Approved}) = \frac{4}{7} = 0.571$$

$$P(\text{Low} | \text{Denied}) = \frac{5}{8} = 0.625$$

$$P(\text{Med} | \text{Denied}) = \frac{3}{8} = 0.375$$

$$P(\text{High} | \text{Denied}) = \frac{0}{8} = 0$$

Credit History:

$$P(\text{Bad} | \text{Approved}) = \frac{0}{7} = 0$$

$$P(\text{Fair} | \text{Approved}) = \frac{3}{7} = 0.429$$

$$P(\text{Good} | \text{Approved}) = \frac{4}{7} = 0.571$$

$$P(\text{Bad} | \text{Denied}) = \frac{4}{8} = 0.5$$

$$P(\text{Fair} | \text{Denied}) = \frac{3}{8} = 0.375$$

$$P(\text{Good} | \text{Denied}) = \frac{1}{8} = 0.125$$

Home Ownership:

$$P(0 | \text{Approved}) = \frac{2}{7} = 0.286$$

$$P(1 | \text{Approved}) = \frac{5}{7} = 0.714$$

$$P(0 | \text{Denied}) = \frac{5}{8} = 0.625$$

$$P(1 | \text{Denied}) = \frac{3}{8} = 0.375$$

Employment Type

$$P(\text{FT} | \text{Approved}) = \frac{5}{7} = 0.7143$$

$$P(\text{PT} | \text{Approved}) = \frac{2}{7} = 0.2857$$

$$P(\text{FT} | \text{Denied}) = \frac{2}{8} = 0.25$$

$$P(\text{PT} | \text{Denied}) = \frac{6}{8} = 0.75$$

d) $P(\text{Med} \cap \text{Fair} \cap 1 \cap \text{FT} | \text{Approved})$

$$= P(\text{Med} | \text{Approved}) \cdot P(\text{Fair} | \text{Approved}) \cdot P(1 | \text{Approved}) \cdot P(\text{FT} | \text{Approved})$$

$$= \frac{3}{7} \times \frac{3}{7} \times \frac{5}{7} \times \frac{5}{7} = \frac{225}{2401}$$

$$\therefore P(\text{Approved} | \text{Med} \cap \text{Fair} \cap 1 \cap \text{FT}) = \frac{225}{2401} \times P(\text{Approved})$$

$$= 0.0437$$

$$P(\text{Denied} | \text{Med} \cap \text{Fair} \cap 1 \cap \text{FT}) = P(\text{Med} \cap \text{Fair} \cap 1 \cap \text{FT} | \text{Denied}) \times P(\text{Den})$$

$$= P(\text{Med} | \text{Denied}) \times P(\text{Fair} | \text{Denied}) \times$$

$$P(1 | \text{Denied}) \times P(\text{FT} | \text{Denied}) \times P(\text{Denied})$$

$$= \frac{3}{8} \times \frac{3}{8} \times \frac{3}{8} \times \frac{8}{15} \times \frac{2}{8}$$

$$= 0.007$$

So, the Application will be Approved.